Chapter 08 Part 2: Enthalpy Change

1. $AgNO_3(aq) + Na\,Br(aq) \rightarrow AgBr\,(s) + NaNO_3(aq)$

A $10.0\ \text{mL}$ sample of $1.0\ M\ \ AgNO_3(aq)$ and a $10.0\ \text{mL}$ sample of $1.0\ M\ \ NaBr(aq)$, both originally at $25^\circ C$, are combined in a constant-pressure calorimeter where the reaction represented above occurs. If the final temperature of the mixture is $35^\circ C$, what is the value of ΔH for the reaction? (Assume that the heat absorbed by the calorimeter is negligible, that the total mass of the mixture is $20.0\ \text{g}$, and that the specific heat capacity of the mixture is $4.2\ \text{J}/(\text{g}\cdot{}^\circ\text{C})$.)

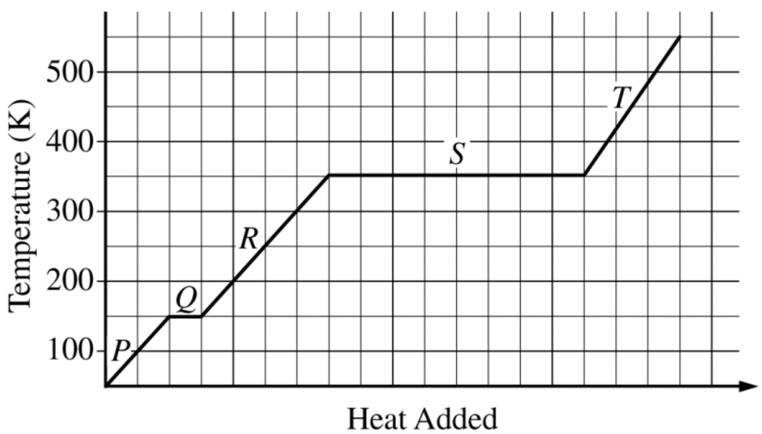
(A) $-84{,}000\ \text{J}/\ \text{mol}_{rxn}$

(B) $-84\ \text{J}/\ \text{mol}_{rxn}$

(C) $+84\ \text{J}/\ \text{mol}_{rxn}$

(D) $+84{,}000\ \text{J}/\ \text{mol}_{rxn}$

2.



The heating curve for a sample of pure ethanol is provided above. The temperature was recorded as a 50.0 g sample of solid ethanol was heated at a constant rate. Which of the following explains why the slope of segment T is greater than the slope of segment R?

(A) The specific heat capacity of the gaseous ethanol is less than the specific heat capacity of liquid ethanol.

(B) The specific heat capacity of the gaseous ethanol is greater than the specific heat capacity of liquid ethanol.

(C) The heat of vaporization of ethanol is less than the heat of fusion of ethanol.

(D) The heat of vaporization of ethanol is greater than the heat of fusion of ethanol.

3. Which of the following phase changes involves the transfer of heat from the surroundings to the system?

(A) $CH_4(g) \rightarrow CH_4(l)$, because CH_4 molecules in the gas phase must absorb energy in order to move closer together, thereby increasing the intermolecular attractions in the solid state.

(B) $CO_2(g) \rightarrow CO_2(s)$, because CO_2 molecules in the gas phase must absorb energy in order to move closer together, thereby increasing the intermolecular attractions in the liquid state.

(C) $H_2O(l) \rightarrow H_2O(s)$, because H_2O molecules in the liquid phase must absorb energy in order to create a crystalline structure with strong intermolecular attractions in the solid state.

(D) $NH_3(l) \rightarrow NH_3(g)$, because NH_3 molecules in the liquid phase must absorb energy in order to overcome their intermolecular attractions and become free gas molecules.

Chapter 08 Part 2: Enthalpy Change

4. Which of the following best helps to explain why the value of $\Delta H°$ for the dissolving of CaF_2 in water is positive?

(A) $CaF_2(s)$ is insoluble in water.

(B) $CaF_2(s)$ dissolves in water to form $CaF_2(aq)$ particles.

(C) Ca^{2+} ions have very strong ion-ion interactions with F^- ions in the crystal lattice.

(D) Ca^{2+} ions have very strong ion-dipole interactions with water molecules in the solution.

5. $CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(g) \qquad \Delta H^\circ < 0$

Which of the following statements must be true for the combustion of $CH_4(g)$, represented by the equation above?

(A) The bond dissociation energy between O atoms in O_2 is greater than the sum of the bond dissociation energies of the two $O-H$ bonds in H_2O.

(B) More bonds are formed in the products than are broken in the reactants.

(C) The sum of the bond dissociation energies of the product molecules is equal to the sum of the bond dissociation energies of the reactant molecules.

(D) The sum of the bond dissociation energies of the product molecules is greater than the sum of the bond dissociation energies of the reactant molecules.

6.

Bond	Approximate Bond Enthalpy (kJ / mol)
$C-H$	410
$C-F$	490
$H-F$	570
$F-F$	160

Based on the bond enthalpies given in the table above, which of the following is the best estimate of ΔH° for the reaction represented below?

$CH_4 + F_2 \rightarrow CH_3F + HF$

(A) $-650\ \text{kJ} / \text{mol}_{rxn}$

(B) $-490\ \text{kJ} / \text{mol}_{rxn}$

(C) $+490\ \text{kJ} / \text{mol}_{rxn}$

(D) $+650\ \text{kJ} / \text{mol}_{rxn}$

Chapter 08 Part 2: Enthalpy Change

7.

$$3\ C_2H_2(g) \rightarrow C_6H_6(g)$$

What is the standard enthalpy change ΔH^o, for the reaction represented above?

(ΔH^o_f of $C_2H_2(g)$ is 230 kJ mol^{-1};

(ΔH^o_f of $C_6H_6(g)$ is 83 kJ mol^{-1};)

(A) -607 kJ

(B) -147 kJ

(C) -19 kJ

(D) +19 kJ

(E) +773 kJ

8.

Mass of Solid	Mass of Water	ΔT_{water}
1.0 g	2000.0 g	$+\ 2.1^{\circ}\text{C}$

A sample of a solid organic compound is completely combusted in a calorimeter. The heat generated by combustion is transferred to the water, causing the temperature of the water to increase. Based on the data in the table above, which of the following is the best estimate of the heat of combustion, ΔH_{comb}, of the organic solid? (The specific heat of water is $4.2\ \text{J}/(\text{g}\cdot{}^{\circ}\text{C})$)

(A) $+18\ \text{kJ}/\text{g}$

(B) $+8.4\ \text{kJ}/\text{g}$

(C) $-8.4\ \text{kJ}/\text{g}$

(D) $-18\ \text{kJ}/\text{g}$

Chapter 08 Part 2: Enthalpy Change

9.

Data for H_2O	
Specific heat capacity of liquid phase	$4.2\ J/(g \cdot {}^\circ C)$
Heat of fusion	$330\ J/g$

A sample of ice at $0\,^\circ C$ is added to $100.$ g of water at $33\,^\circ C$. The mixture is stirred gently until the temperature of the water is $0\,^\circ C$. All the remaining ice is quickly removed. Based on the information in the table above, the mass of ice that melted is closest to

(A) 0.13 g

(B) 0.42 g

(C) 1.3 g

(D) 42 g

$$CH_3OH(g) \rightarrow CO(g) + 2\ H_2(g) \qquad \Delta H^\circ = +91\ kJ/mol_{rxn}$$

The reaction represented above goes essentially to completion. The reaction takes place in a rigid, insulated vessel that is initially at 600 K

10. Which of the following statements about the bonds in the reactants and products is most accurate?

(A) The sum of the bond enthalpies of the bonds in the reactant is greater than the sum of the bond enthalpies of the bonds in the products.

(B) The sum of the bond enthalpies of the bonds in the reactant is less than the sum of the bond enthalpies of the bonds in the products.

(C) The length of the bond between carbon and oxygen in CH_3OH is shorter than the length of the bond between carbon and oxygen in CO.

(D) All of the bonds in the reactant and products are polar.

11.

$$CO(g) + 2\ H_2(g) \rightleftarrows CH_3OH(g) \qquad \Delta H < 0$$

The synthesis of $CH_3OH(g)$ from $CO(g)$ and $H_2(g)$ is represented by the equation above. The value of K_c for the reaction at 483 K is 14.5.

Which of the following statements is true about bond energies in this reaction?

Chapter 08 Part 2: Enthalpy Change

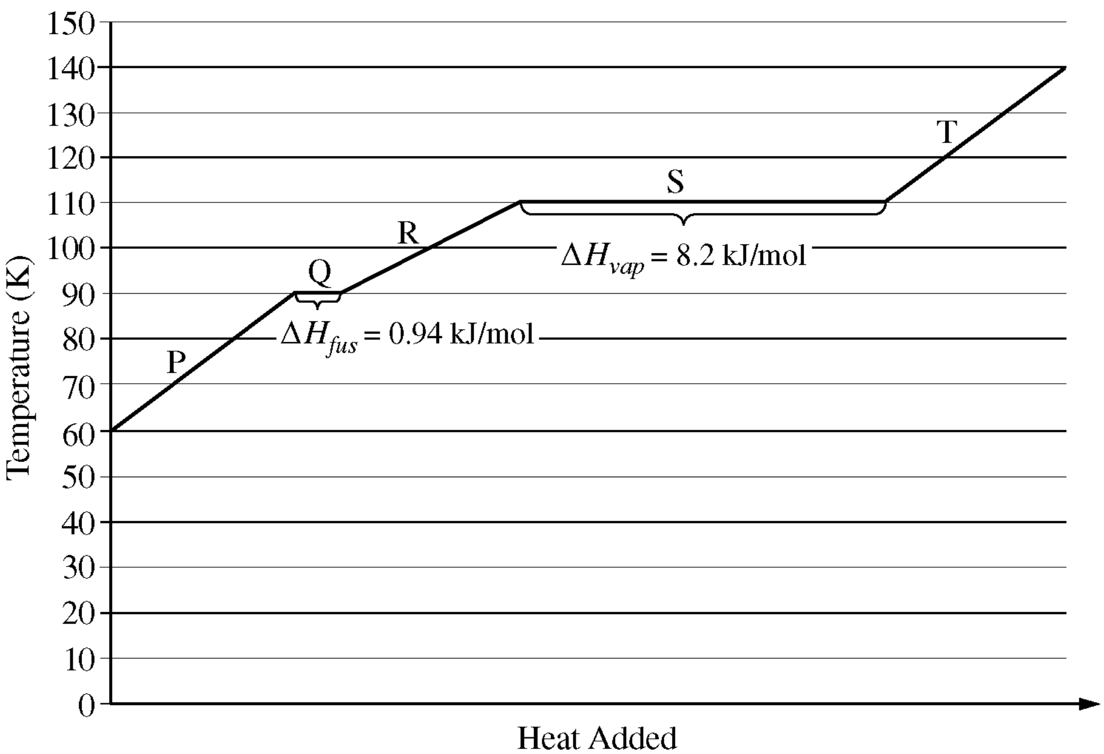
(A) The energy absorbed as the bonds in the reactants are broken is greater than the energy released as the bonds in the product are formed.

(B) The energy released as the bonds in the reactants are broken is greater than the energy absorbed as the bonds in the product are formed.

(C) The energy absorbed as the bonds in the reactants are broken is less than the energy released as the bonds in the product are formed.

(D) The energy released as the bonds in the reactants are broken is less than the energy absorbed as the bonds in the product are formed.

The following questions refer to the graph below, which shows the heating curve for methane, CH_4.



12. How much energy is required to melt 64 g of methane at 90 K? (The molar mass of methane is 16 g/mol.)

(A) 0.24 kJ

(B) 3.8 kJ

(C) 33 kJ

(D) 60. kJ

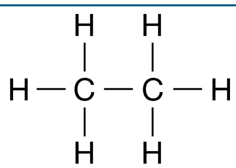
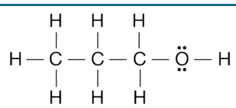
13. The enthalpy of vaporization of water is 40.7 kJ/mol. Which of the following best explains why the enthalpy of vaporization of methane is less than that of water?

Chapter 08 Part 2: Enthalpy Change

(A) Methane does not exhibit hydrogen bonding, but water does.
(B) Methane has weaker dispersion forces.
(C) Methane has a smaller molar mass.
(D) Methane has a much lower density.

14. Which of the following best explains why more energy is required for the process occurring at 110 K than for the process occurring at 90 K ?

(A) Intermolecular attractions are completely overcome during vaporization.
(B) Intermolecular attractions in the solid phase are weaker than in the liquid phase.
(C) Electron clouds of methane molecules are less polarizable at lower temperatures.
(D) Vaporization involves a large increase in temperature.

Compound	**Molar Mass**	**Balanced Equation for the Combustion of One Mole of the Compound**	ΔH°_{comb} $(\mathrm{kJ/mol}_{rxn})$
 Ethane	$30.\,\mathrm{g/mol}$	$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2\,CO_2(g) + 3\,H_2O(l)$	-1560
 Propanol	$60.\,\mathrm{g/mol}$	$C_3H_8O(l) + \frac{9}{2}O_2(g) \rightarrow 3\,CO_2(g) + 4\,H_2O(l)$	-2020
Unknown	?	$? +\ 5\,O_2(g) \rightarrow 4\,CO_2(g) + 4\,H_2O(l)$	-2240

15. In an experiment, 30.0 g of ethane and 30.0 g of propanol are placed in separate reaction vessels. Each compound undergoes complete combustion with excess $O_2(g)$. Which of the following best compares the quantity of heat released in each combustion reaction?

(A) $q_{ethane} < q_{propanol}$
(B) $q_{ethane} = q_{propanol}$
(C) $q_{ethane} > q_{propanol}$
(D) The quantities of heat released in the combustions of ethane and propanol cannot be compared without knowing the specific heat capacity of the compounds.

16. Which of the following best explains why the combustion reactions represented in the table are exothermic?

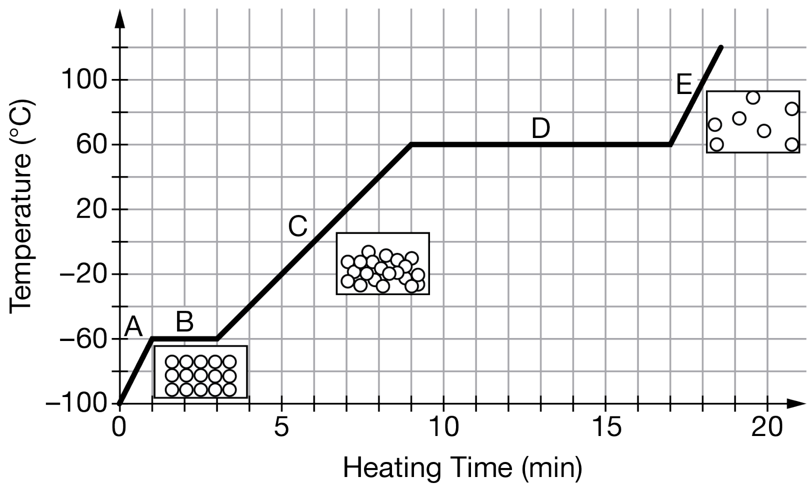
(A) The number of bonds in the reactant molecules is greater than the number of bonds in the product molecules.

(B) The number of bonds in the reactant molecules is less than the number of bonds in the product molecules.

(C) The energy required to break the bonds in the reactants is greater than the energy released in forming the bonds in the products.

(D) The energy required to break the bonds in the reactants is less than the energy released in forming the bonds in the products.

17.



A sample of $CHCl_3(s)$ was exposed to a constant source of heat for a period of time. The graph above shows the change in the temperature of the sample as heat is added. Which of the following best describes what occurs at the particle level that makes segment D longer than segment B?

(A) The specific heat capacity of the liquid is significantly higher than that of the solid, because the particles in the liquid state need to absorb more thermal energy to increase their average speed.

(B) The specific heat capacity of the solid is significantly higher than that of the gas, because the particles in the solid state need to absorb more thermal energy to increase their average speed.

(C) The enthalpy of fusion is greater than the enthalpy of vaporization, because separating molecules from their bound crystalline state requires more energy than separating molecules completely from the liquid state.

(D) The enthalpy of vaporization is greater than the enthalpy of fusion, because separating molecules completely from the liquid to form a gas requires more energy than separating molecules from their bound crystalline state to a liquid state.

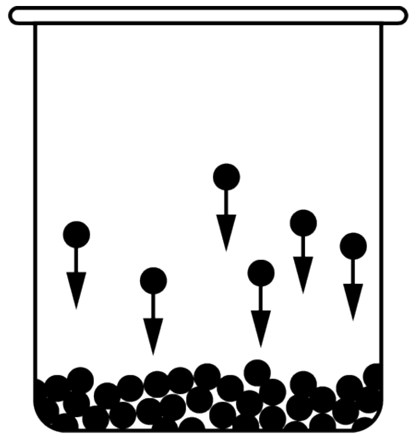
$2\ \mathrm{K}(s) + 2\ \mathrm{H_2O}(l) \rightarrow 2\ \mathrm{KOH}(aq) + \mathrm{H_2}(g) \qquad \Delta H^\circ = -393\ \mathrm{kJ/mol}_{rxn}$

When 3.9 g of $\mathrm{K}(s)$ is added to $200.$ mL of water at $25^\circ\mathrm{C}$ in a calorimeter, all the $\mathrm{K}(s)$ reacts with the water as represented by the equation above.

18. Which of the following is true when the 3.9 g of $\mathrm{K}(s)$ has completely reacted?

(A) $10.$ kJ of energy is absorbed by the reaction.
(B) $20.$ kJ of energy is absorbed by the reaction.
(C) $10.$ kJ of energy is released by the reaction.
(D) $20.$ kJ of energy is released by the reaction.

19.



A 2.00 mol sample of C_2H_5OH undergoes the phase transition illustrated in the diagram above. The molar enthalpy of vaporization, ΔH_{vap}, of C_2H_5OH is $+38.6$ kJ/mol. Which of the following best identifies the change in enthalpy in the phase transition shown in the diagram?

(A) $+19.3$ kJ
(B) $+77.2$ kJ
(C) -19.3 kJ
(D) -77.2 kJ

20.

Bond Type	Average Bond Enthalpy $\left(\frac{kJ}{mol}\right)$
$C-C$	360
$C=O$	799
$C\equiv O$	1072
$O-O$	142
$O=O$	498

The oxidation of carbon monoxide can be represented by the chemical equation $2\,CO(g) + O_2(g) \rightarrow 2\,CO_2(g)$. The table above provides the average bond enthalpies for different bond types. Based on the information in the table, which of the following mathematical expressions is correct for the estimated enthalpy change for the reaction?

(A) $\Delta H_{rxn} = [2(1072\frac{\text{kJ}}{\text{mol}}) + (498\ \frac{\text{kJ}}{\text{mol}})] - 2(799\ \frac{\text{kJ}}{\text{mol}})$

(B) $\Delta H_{rxn} = [2(1072\frac{\text{kJ}}{\text{mol}}) + (498\ \frac{\text{kJ}}{\text{mol}})] - 4(799\ \frac{\text{kJ}}{\text{mol}})$.

(C) $\Delta H_{rxn} = [2(799\ \frac{\text{kJ}}{\text{mol}}) + (142\frac{\text{kJ}}{\text{mol}})] - 4(360\ \frac{\text{kJ}}{\text{mol}})$

(D) $\Delta H_{rxn} = [2(799\ \frac{\text{kJ}}{\text{mol}}) + (142\frac{\text{kJ}}{\text{mol}})] - 2(360\ \frac{\text{kJ}}{\text{mol}})$

21.

Reaction 1:	$\text{N}_2\text{O}_4(g) \rightarrow 2\ \text{NO}_2(g)$	$\Delta H_1 = +57.9\ \text{kJ}$
Reaction 2:	$2\ \text{NO}(g) + \text{O}_2(g) \rightarrow 2\ \text{NO}_2(g)$	$\Delta H_2 = -113.1\ \text{kJ}$

Based on the information for two different reactions given above, which of the following gives the quantities needed to calculate the enthalpy change for the reaction represented by the overall equation below?

$2\ \text{NO}(g) + \text{O}_2(g) \rightarrow \text{N}_2\text{O}_4(g)$

(A) $\Delta H_1 + \Delta H_2$

(B) $\Delta H_1 + (-\Delta H_2)$

(C) $(-\Delta H_1) + \Delta H_2$

(D) $\Delta H_1 + (2 \times \Delta H_2)$

22. $AgNO_3(aq)\ +\ NaC1(aq) \rightarrow\ AgC1(s)\ +\ NaNO_3(aq)$

In an experiment a student mixes a 50.0 mL sample of 0.100 *M* $AgNO_3(aq)$ with a 50.0 mL sample of 0.100 *M* NaCl(*aq*) at 20.0°C in a coffee-cup calorimeter. Which of the following is the enthalpy change of the precipitation reaction represented above if the final temperature of the mixture is 21.0°C? (Assume that the total mass of the mixture is 100. g and that the specific heat capacity of the mixture is 4.2 J/(g °C).)

(A) -84 kJ/mol$_{rxn}$

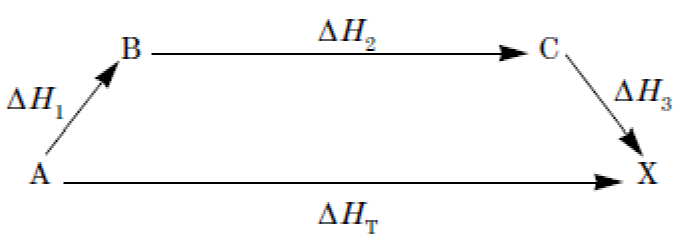
(B) -0.42 kJ/mol$_{rxn}$

(C) 0.42 kJ/mol$_{rxn}$

(D) 84 kJ/mol$_{rxn}$

23. A → X

The enthalpy change for the reaction represented above is ΔH_T. This reaction can be broken down into a series of steps as shown in the diagram:



A relationship that must exist among the various enthalpy changes is

Chapter 08 Part 2: Enthalpy Change

(A) $\Delta H_T - \Delta H_1 - \Delta H_2 - \Delta H_3 = 0$
(B) $\Delta H_T + \Delta H_1 + \Delta H_2 + \Delta H_3 = 0$
(C) $\Delta H_3 - (\Delta H_1 + \Delta H_2) = \Delta H_T$
(D) $\Delta H_2 - (\Delta H_3 + \Delta H_1) = \Delta H_T$
(E) $\Delta H_T + \Delta H_2 = \Delta H_1 + \Delta H_3$

24. $2\ H_2O_2(l) \rightarrow 2\ H_2O(l) + O_2(g)$

Bond	Average Bond Enthalpy (kJ/mol)
$O-H$	463
$O-O$	146
$O{=}O$	495

Shown above are the equation representing the decomposition of $H_2O_2(l)$ and a table of bond enthalpies. On the basis of the information, which of the following is the enthalpy of decomposition of 2 mol of $H_2O_2(l)$?

(A) -349 kJ
(B) -203 kJ
(C) 203 kJ
(D) 349 kJ

25. $O_2(g) \rightarrow 2\ O(g)$

Bond	Average Bond Enthalpy (kJ/mol)
$O{=}O$	495
$O-O$	146

An equation representing the dissociation of $O_2(g)$ and a table of bond enthalpies are shown above. Based on the information, which of the following is the enthalpy of dissociation for $O_2(g)$?

(A) -641 kJ/mol
(B) -495 kJ/mol
(C) 495 kJ/mol
(D) 641 kJ/mol

26.

$$2\ NH_3(g) \rightarrow 3\ H_2(g) + N_2(g)\ \Delta H^\circ_{298} = 92\ \text{kJ/mol}_{rxn}$$

According to the information above, what is the standard enthalpy of formation, ΔH°_f , for $NH_3(g)$ at 298 K ?

(A) -92 kJ/mol
(B) -46 kJ/mol
(C) 46 kJ/mol
(D) 92 kJ/mol
(E) 184 kJ/mol

27.

Substance	ΔH_f° (kJ/mol)
$Fe_2O_3(s)$	-826
$CO(g)$	-111
$Fe(s)$	0
$CO_2(g)$	-394

Based on the information in the table above, which of the following expressions gives the approximate ΔH° for the reaction represented by the following balanced chemical equation?

$Fe_2O_3(s) + 3\ CO(g) \rightarrow 2\ Fe(s) + 3\ CO_2(g)$

(A) $\Delta H^\circ{}_{rxn} = [(0\text{ kJ/mol}) + (-394\text{ kJ/mol})] - [(-826\text{ kJ/mol}) + (-111\text{ kJ/mol})]$
(B) $\Delta H^\circ{}_{rxn} = [2(0\text{ kJ/mol}) + 3(-394\text{ kJ/mol})] - [(-826\text{ kJ/mol}) + 3(-111\text{ kJ/mol})]$
(C) $\Delta H^\circ{}_{rxn} = [(-826\text{ kJ/mol}) + 3(-111\text{ kJ/mol})] - [2(0\text{ kJ/mol}) + 3(-394\text{ kJ/mol})]$
(D) $\Delta H^\circ{}_{rxn} = [(-826\text{ kJ/mol}) + (-111\text{ kJ/mol})] - [(0\text{ kJ/mol}) + (-394\text{ kJ/mol})]$

28.

$$\tfrac{1}{2}\ H_2(g) + \tfrac{1}{2}\ I_2(s) \rightarrow HI(g) \qquad \Delta H = 26\text{ kJ/mol}_{rxn}$$

$$\tfrac{1}{2}\ H_2(g) + \tfrac{1}{2}\ I_2(g) \rightarrow HI(g) \qquad \Delta H = -5.0\text{ kJ/mol}_{rxn}$$

Based on the information above, what is the enthalpy change for the sublimation of iodine, represented below?

$$I_2(s) \rightarrow I_2(g)$$

(A) 15 kJ/mol_{rxn}
(B) 21 kJ/mol_{rxn}
(C) 31 kJ/mol_{rxn}
(D) 42 kJ/mol_{rxn}
(E) 62 kJ/mol_{rxn}

Chapter 08 Part 2: Enthalpy Change

29.

Substance	ΔH_f° (kJ/mol)
$Al(s)$	0
$Al_2O_3(s)$	-1680
$Fe(s)$	0
$Fe_2O_3(s)$	?

The enthalpy change for the reaction $2\ Al(s) + Fe_2O_3(s) \rightarrow 2\ Fe(s) + Al_2O_3(s)$ is $-860\ kJ/mol$. Based on the standard enthalpies of formation ΔH_f° provided in the table, what is the approximate ΔH_f° for $Fe_2O_3(s)$?

(A) $+2540\ kJ/mol$
(B) $-2540\ kJ/mol$
(C) $+820\ kJ/mol$
(D) $-820\ kJ/mol$

30. $C(s) + H_2O(g) \rightarrow CO(g) + H_2(g) \qquad \Delta H^\circ = +131\ kJ/mol_{rxn}$

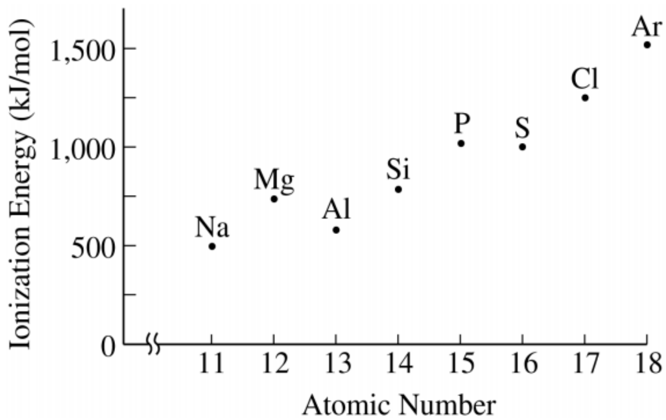
The reaction between $C(s)$ and $H_2O(g)$ is represented by the balanced chemical equation above. Based on the enthalpy change of the reaction (ΔH°) and the standard heats of formation (ΔH_f°) given in the table below, what is the approximate ΔH_f° for $CO(g)$?

Substance	ΔH_f° **(kJ/mol)**
$C(s)$	0
$H_2O(g)$	-242
$CO(g)$	?
$H_2(g)$	0

(A) $-373\ kJ/mol$
(B) $-111\ kJ/mol$
(C) $+111\ kJ/mol$
(D) $+373\ kJ/mol$

Chapter 08 Part 2: Enthalpy Change

31.



The first ionization energy of an element is the energy required to remove an electron from a gaseous atom of the element (i.e., $X(g) \rightarrow X^{+}(g) + e^{-}$). The values of the first ionization energies for the third-row elements are shown in the graph above. On the basis of the information given, which of the following reactions is exothermic?

(A) $Cl(g) + Mg^{+}(g) \rightarrow Cl^{+}(g) + Mg(g)$

(B) $Al(g) + Mg^{+}(g) \rightarrow Al^{+}(g) + Mg(g)$

(C) $P(g) + Mg^{+}(g) \rightarrow P^{+}(g) + Mg(g)$

(D) $S(g) + Mg^{+}(g) \rightarrow S^{+}(g) + Mg(g)$

32. $CH_4(g) + Cl(g) \rightarrow CH_3(g) + HCl(g)$ $\Delta H° = -14$ kJ/mol$_{rxn}$

$NH_3(g) + Cl(g) \rightarrow NH_2(g) + HCl(g)$ $\Delta H° = -36$ kJ/mol$_{rxn}$

$H_2O(g) + Cl(g) \rightarrow OH(g) + HCl(g)$ $\Delta H° = +40$ kJ/mol$_{rxn}$

Based on the data above, what can be concluded regarding the strength of the C-H, N-H, and O-H bonds in the molecules shown?

(A) The C-H bond is the strongest.

(B) The N-H bond is the strongest.

(C) The O-H bond is the strongest.

(D) Nothing can be concluded without knowing the strength of the H- Cl bond.

Chapter 08 Part 2: Enthalpy Change

$$2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g) \qquad \Delta H° = -196\ \text{kJ/mol}_{rxn}$$

The decomposition of $H_2O_2(aq)$ is represented by the equation above. A student monitored the decomposition of a 1.0 L sample of $H_2O_2(aq)$ at a constant temperature of 300. K and recorded the concentration of H_2O_2 as a function of time. The results are given in the table below.

Time (s)	$[H_2O_2]$
0	2.7
200.	2.1
400.	1.7
600.	1.4

33. Assume that the bond enthalpies of the oxygenhydrogen bonds in H_2O are not significantly different from those in H_2O_2 . Based on the value of ΔH° of the reaction, which of the following could be the bond enthalpies (in kJ/mol) for the bonds broken and formed in the reaction?

Chapter 08 Part 2: Enthalpy Change

(A)

O–O in H2O2	O=O in O2	O–H
300	500	500

(B)

O–O in H2O2	O=O in O2	O–H
150	500	500

(C)

O–O in H2O2	O=O in O2	O–H
500	300	150

(D)

O–O in H2O2	O=O in O2	O–H
250	300	150

$$H_2(g) + Cl_2(g) \rightleftarrows 2\,HCl(g) \qquad K_p = 2 \times 10^{30} \text{ at } 298 \text{ K}$$

$HCl(g)$ can be synthesized from $H_2(g)$ and $Cl_2(g)$ as represented above. A student studying the kinetics of the reaction proposes the following mechanism

Step 1: $Cl_2(g) \rightarrow 2\,Cl(g)$ (*slow*) $\Delta H° = 242$ kJ/mol$_{rxn}$
Step 2: $H_2(g) + Cl(g) \rightarrow HCl(g) + H(g)$ (*fast*) $\Delta H° = 4$ kJ/mol$_{rxn}$
Step 3: $H(g) + Cl(g) \rightarrow HCl(g)$ (*fast*) $\Delta H° = -432$ kJ/mol$_{rxn}$

34. What is the value of the enthalpy change per mole of $HCl(g)$ produced?

(A) -93 kJ
(B) -121 kJ
(C) -186 kJ
(D) -242 kJ

35.

(1)	$2\ \mathrm{H}(g) \rightarrow \mathrm{H_2}(g)$	$\Delta H^\circ{}_1 = -436$ kJ
(2)	$2\ \mathrm{O}(g) \rightarrow \mathrm{O_2}(g)$	$\Delta H^\circ{}_2 = -249$ kJ
(3)	$2\ \mathrm{H}(g) + \mathrm{O}(g) \rightarrow \mathrm{H_2O}(g)$	$\Delta H^\circ{}_3 = -803$ kJ
(4)	$\mathrm{H_2O}(g) \rightarrow \mathrm{H_2O}(l)$	$\Delta H^\circ{}_4 = -44$ kJ
(5)	$\mathrm{H_2}(g) + \frac{1}{2}\ \mathrm{O_2}(g) \rightarrow \mathrm{H_2O}(l)$	$\Delta H_f^\circ = ?$

Based on the chemical equations and their associated enthalpy changes shown above, which of the following identifies the quantities needed to calculate ΔH_f°, the standard enthalpy of formation of $\mathrm{H_2O}(l)$, in $\mathrm{kJ/mol}$?

(A) $2(-\Delta H^\circ{}_1)+(-\Delta H^\circ{}_2) + 2(\Delta H^\circ{}_3) + 2(\Delta H^\circ{}_4)$
(B) $(-\Delta H^\circ{}_1) + \frac{1}{2}(-\Delta H^\circ{}_2) + (\Delta H^\circ{}_3) + (\Delta H^\circ{}_4)$
(C) $(-\Delta H^\circ{}_1) + \frac{1}{2}(\Delta H^\circ{}_2)+(\Delta H^\circ{}_3)$
(D) $(\Delta H^\circ{}_1) + \frac{1}{2}(\Delta H^\circ{}_2) + (\Delta H^\circ{}_3) + (\Delta H^\circ{}_4)$

36. $2\ \mathrm{H_2S}(g) + 3\ \mathrm{O_2}(g) \rightarrow 2\ \mathrm{H_2O}(l) + 2\ \mathrm{SO_2}(g) \qquad \Delta H^\circ = -1120\ \mathrm{kJ/mol}_{rxn}$

Based on the reaction represented by the chemical equation shown above, what is the amount of heat released when 4.00 mol of $\mathrm{H_2S}(g)$ reacts with 9.00 mol of $\mathrm{O_2}(g)$?

(A) -560 kJ
(B) -1120 kJ
(C) -2240 kJ
(D) -3360 kJ

37. $\mathrm{CS_2}(l) + 2\ \mathrm{H_2O}(l) \rightarrow \mathrm{CO_2}(g) + 2\ \mathrm{H_2S}(g) \quad \Delta H^\circ{}_{\mathrm{rxn}} = ?$

Which of the following combinations represents the individual reactions and the quantities needed to determine ΔH° for the overall reaction represented by the chemical equation above?

Chapter 08 Part 2: Enthalpy Change

(A)

$$CS_2(l) + 3\ O_2(g) \rightarrow CO_2(g) + 2\ SO_2(g) \qquad \Delta H^\circ = -1075\ \text{kJ}$$

$$2\ H_2O(l) + 2\ SO_2(g) \rightarrow 2\ H_2S(g) + 3\ O_2(g) \qquad \Delta H^\circ = +1136\ \text{kJ}$$

(B)

$$CS_2(l) + 3\ O_2(g) \rightarrow CO_2(g) + 2\ SO_2(g) \qquad \Delta H^\circ = -1075\ \text{kJ}$$

$$H_2O(l) + SO_2(g) \rightarrow H_2S(g) + \tfrac{3}{2}\ O_2(g) \qquad \Delta H^\circ = +568\ \text{kJ}$$

(C)

$$CS_2(l) + 3\ O_2(g) \rightarrow CO_2(g) + 2\ SO_2(g) \qquad \Delta H^\circ = -1075\ \text{kJ}$$

$$H_2S(g) + \tfrac{3}{2}\ O_2(g) \rightarrow H_2O(l) + SO_2(g) \qquad \Delta H^\circ = -568\ \text{kJ}$$

(D)

$$CS_2(l) + 3\ O_2(g) \rightarrow CO_2(g) + 2\ SO_2(g) \qquad \Delta H^\circ = -1075\ \text{kJ}$$

$$2\ H_2S(g) + 3\ O_2(g) \rightarrow 2\ H_2O(l) + 2\ SO_2(g) \qquad \Delta H^\circ = -1136\ \text{kJ}$$

38. $Mg(s) + 2\ HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

The chemical equation shown above represents the reaction between $Mg(s)$ and $HCl(aq)$. When 12.15 g of $Mg(s)$ is added to 500.0 mL of $4.0\ M\ HCl(aq)$, 95 kJ of heat is released. The experiment is repeated with 24.30 g of $Mg(s)$ and 500.0 mL of $4.0\ M\ HCl(aq)$. Which of the following gives the correct value for the amount of heat released by the reaction?

(A) 380 kJ
(B) 190 kJ
(C) 95 kJ
(D) 48 kJ

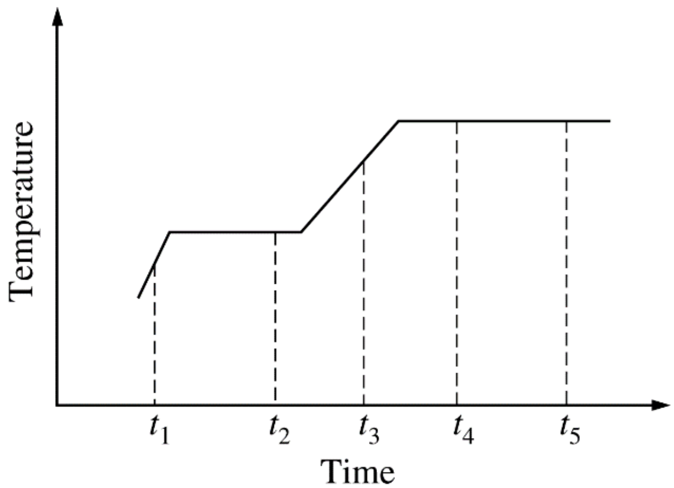
39. $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l) \qquad \Delta H^\circ = -57.1\ \text{kJ/mol}_{rxn}$

The chemical equation above represents the reaction between $HCl(aq)$ and $NaOH(aq)$. When equal volumes of $1.00\ M\ HCl(aq)$ and $1.00\ M\ NaOH(aq)$ are mixed, 57.1 kJ of heat is released. If the experiment is repeated with $2.00\ M\ HCl(aq)$, how much heat would be released?

Chapter 08 Part 2: Enthalpy Change

(A) 28.6 kJ
(B) 57.1 kJ
(C) 85.7 kJ
(D) 114 kJ

The following questions relate to the graph below. The graph shows the temperature of a pure substance as it is heated at a constant rate in an open vessel at 1.0 atm pressure. The substance changes from the solid to the liquid to the gas phase.



40. The substance is at its normal freezing point at time

(A) t_1
(B) t_2
(C) t_3
(D) t_4
(E) t_5

41. Which of the following best describes what happens to the substance between t_4 and t_5?

(A) The molecules are leaving the liquid phase.
(B) The solid and liquid phases coexist in equilibrium.
(C) The vapor pressure of the substance is decreasing.
(D) The average intermolecular distance is decreasing.
(E) The temperature of the substance is increasing.

Chapter 08 Part 2: Enthalpy Change

$K(s) + \frac{1}{2}\ Cl_2(g) \rightarrow KCl(s)$ $\Delta H° = -437 kJ/mol_{rxn}$

The elements K and Cl react directly to form the compound KCl according to the equation above. Refer to the information above and the table below to answer the questions that follow.

Process	$\Delta H°$ (kJ/mol_{rxn})
$K(s) \rightarrow K(g)$	v
$K(g) \rightarrow K^+(g) + e^-$	w
$Cl_2(g) \rightarrow 2\ Cl(g)$	x
$Cl(g) + e^- \rightarrow Cl^-(g)$	y
$K^+(g) + Cl^-(g) \rightarrow KCl(s)$	z

42. $Cl_2(g) + 2e^- \rightarrow 2Cl^-(g)$

Which of the following expressions is equivalent to ΔH^o for the reaction represented above?

(A) $x + y$

(B) $x - y$

(C) $x + 2y$

(D) $\frac{x}{2} - y$

43. How much heat is released or absorbed when 0.050 mol of $Cl_2(g)$ is formed from $KCl(s)$?

(A) 87.4 kJ is released

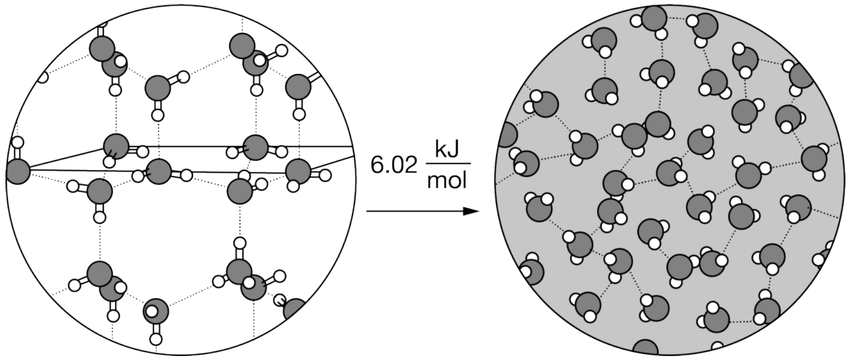
(B) 43.7 kJ is released

(C) 43.7 kJ is absorbed

(D) 87.4 kJ is absorbed

Chapter 08 Part 2: Enthalpy Change

44.



The diagram above represents the melting of $H_2O(s)$. A 2.00 mole sample of $H_2O(s)$ at $0^\circ C$ melted, producing $H_2O(l)$ at $0^\circ C$. Based on the diagram, which of the following best describes the amount of heat required for this process and the changes that took place at the molecular level?

(A) 3.01 kJ of heat was absorbed to decrease the average speed of the water molecules in the liquid, which decreases the distance between molecules.

(B) 6.02 kJ of heat was absorbed to increase the number of hydrogen bonds between water molecules in the liquid compared to the solid.

(C) 12.0 kJ of heat was absorbed to decrease the polarity of the water molecules, which increases the density of the liquid compared to the solid.

(D) 12.0 kJ of heat was absorbed to overcome some of the hydrogen bonding forces holding the water molecules in fixed positions in the crystalline structure.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g) \quad K_p = 3.0 \text{ at } 70°C$$

colorless brown

A mixture of $NO_2(g)$ and $N_2O_4(g)$ is placed in a glass tube and allowed to reach equilibrium at 70°C, as represented above.

45. Which of the following statements about $\Delta H°$ for the reaction is correct?

(A) $\Delta H° < 0$ because energy is released when the N−N bond breaks.

(B) $\Delta H° < 0$ because energy is required to break the N−N bond.

(C) $\Delta H° > 0$ because energy is released when the N−N bond breaks.

(D) $\Delta H° > 0$ because energy is required to break the N−N bond.

Chapter 08 Part 2: Enthalpy Change

When water is added to a mixture of $Na_2O_2(s)$ and S (s) , a redox reaction occurs, as represented by the equation below.

$$2\ Na_2O_2(s) + S(s) + 2\ H_2O(l) \rightarrow 4\ NaOH(aq) + SO_2(aq)$$

$$\Delta H^\circ_{298} = -610\ \text{kJ/mol}_{rxn}\ ;\ \Delta S^\circ_{298} = -7.3\ \text{J/(K}\cdot\text{mol}_{rxn})$$

46. Two trials are run, using excess water. In the first trial, 7.8 g of $Na_2O_2(s)$ (molar mass 78 g/mol) is mixed with 3.2 g of $S(s)$. In the second trial, 7.8 g of $Na_2O_2(s)$ is mixed with 6.4 g of $S(s)$. The $Na_2O_2(s)$ and $S(s)$ react as completely as possible. Both trials yield the same amount of $SO_2(aq)$. Which of the following identifies the limiting reactant and the heat released, q, for the two trials at 298 K?

(A)

Limiting Reactant	q
S	30. kJ

(B)

Limiting Reactant	q
S	61 kJ

(C)

Limiting Reactant	q
Na_2O_2	30. kJ

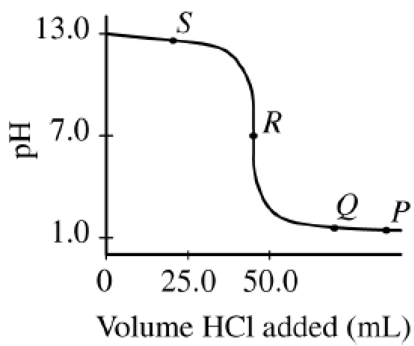
(D)

Limiting Reactant	q
Na_2O_2	61 kJ

Chapter 08 Part 2: Enthalpy Change

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

To determine the concentration of a NaOH(*aq*) solution, a student titrated a 50. mL sample with 0.10 *M* HCl(*aq*) . The reaction is represented by the equation above. The titration is monitored using a pH meter, and the experimental results are plotted in the graph below.



47. A student mixes a 10.0 mL sample of 1.0 M NaOH(*aq*) with a 10.0 mL sample of 1.0 M HCl(*aq*) in a polystyrene container. The temperature of the solutions before mixing was 20.0°C. If the final temperature of the mixture is 26.0°C, what is the experimental value of ΔH°_{rxn} ? (Assume that the solution mixture has a specific heat of 4.2 J/(g•K) and a density of 1.0 g/mL.)

(A) -50. kJ/mol_{rxn}
(B) -25 kJ/mol_{rxn}
(C) -5.0×10^4 kJ/mol_{rxn}
(D) -5.0×10^2 kJ/mol_{rxn}

---

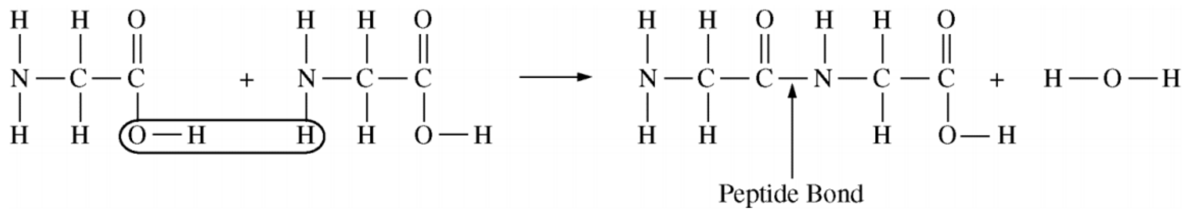
48. $4\ NH_3(g) + 3\ O_2(g) \rightarrow 2\ N_2(g) + 6\ H_2O(g)$

If the standard molar heats of formation of ammonia, $NH_3(g)$, and gaseous water, $H_2O(g)$, are -46 kJ/mol and -242 kJ/mol, respectively, what is the value of ΔH°_{298} for the reaction represented above?

(A) -190 kJ/mol_{rxn}
(B) -290 kJ/mol_{rxn}
(C) -580 kJ/mol_{rxn}
(D) -1,270 kJ/mol_{rxn}
(E) -1,640 kJ/mol_{rxn}

Chapter 08 Part 2: Enthalpy Change

49.



Two molecules of the amino acid glycine join through the formation of a peptide bond, as shown above. The thermodynamic data for the reaction are listed in the following table.

ΔG°_{298}	ΔH°_{298}	ΔS°_{298}
$+15\ \text{kJ/mol}_{rxn}$	$+12\ \text{kJ/mol}_{rxn}$	$-10\ \text{J/(K}\cdot\text{mol}_{rxn})$

Bond	Bond Energy (kJ/mol)
C−O	358
N−H	391
O−H	467

Based on the bond energies listed in the table above, which of the following is closest to the bond energy of the C−N bond?

(A) 197 kJ/mol

(B) 270 kJ/mol

(C) 294 kJ/mol

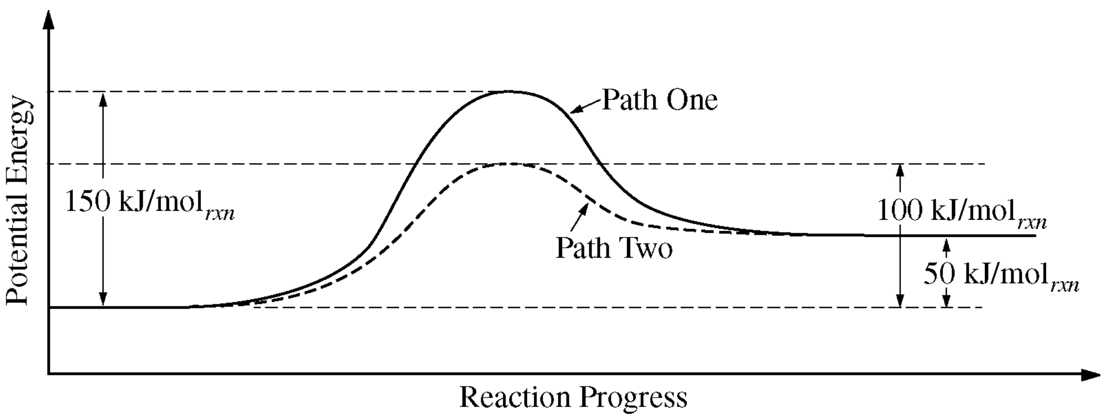
(D) 455 kJ/mol

Chapter 08 Part 2: Enthalpy Change

The following questions relate to the below information.

$$XY_2 \rightarrow X + Y_2$$

The equation above represents the decomposition of a compound XY_2. The diagram below shows two reaction profiles (path one and path two) for the decomposition of XY_2.



50. Which of the following best describes the flow of heat when 1.0 mol of XY_2 decomposes?

(A) 50 kJ of heat is transferred to the surroundings.

(B) 50 kJ of heat is transferred from the surroundings.

(C) 100 kJ of heat is transferred to the surroundings.

(D) 100 kJ of heat is transferred from the surroundings.

51. A molecular solid coexists with its liquid phase at its melting point. The solid-liquid mixture is heated, but the temperature does not change while the solid is melting. The best explanation for this phenomenon is that the heat absorbed by the mixture

(A) is lost to the surroundings very quickly

(B) is used in overcoming the intermolecular attractions in the solid

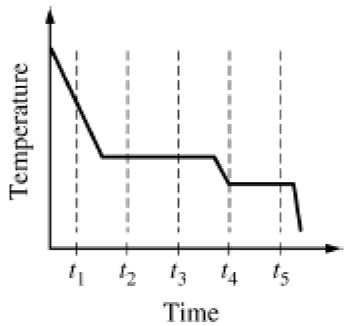
(C) is used in breaking the bonds within the molecules of the solid

(D) causes the nonbonding electrons in the molecules to move to lower energy levels

(E) causes evaporation of the liquid, which has a cooling effect

Chapter 08 Part 2: Enthalpy Change

52.



The cooling curve above shows how the temperature of a sample varies with time as the sample goes through phase changes. The sample starts as a gas, and heat is removed at a constant rate. At which time does the sample contain the most liquid?

(A) t_1

(B) t_2

(C) t_3

(D) t_4

(E) t_5

53. Aluminum metal can be recycled from scrap metal by melting the metal to evaporate impurities.

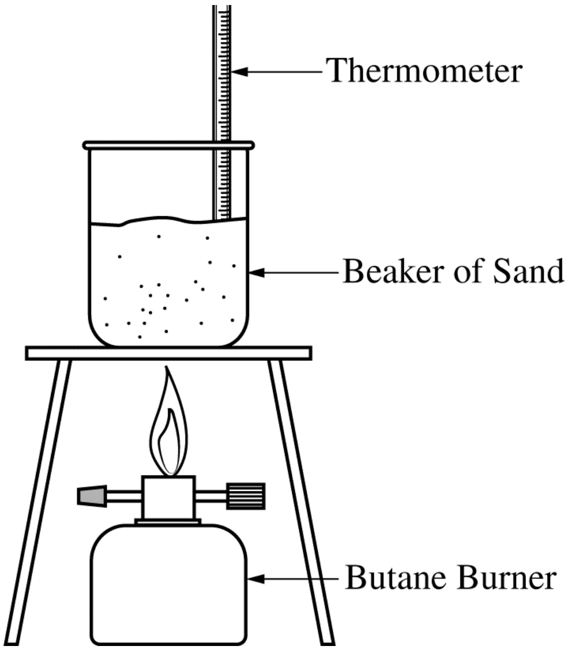
a. Calculate the amount of heat needed to purify 1.00 mole of Al originally at 298 K by melting it. The melting point of Al is 933 K. The molar heat capacity of Al is 24 J/(mol·K), and the heat of fusion of Al is 10.7 kJ/mol.

b. The equation for the overall process of extracting Al from Al_2O_3 is shown below. Which requires less energy, recycling existing Al or extracting Al from Al_2O_3? Justify your answer with a calculation.

$Al_2O_3(s) \rightarrow 2\ Al(s) + \frac{3}{2}\ O_2(g) \qquad \Delta H° = 1675\ kJ/mol_{rxn}$

Chapter 08 Part 2: Enthalpy Change

54. A student is asked to determine what mass of butane, $C_4H_{10}(g)$, needs to burn in order to raise the temperature of a 1650 g beaker of sand by 180.°C. The student is provided with the equipment shown below.



a. Calculate the amount of heat energy needed to raise the temperature of the beaker of sand by 180.°C. Assume that all the heat energy from the burner is transferred to the beaker of sand and that the specific heat capacity of the beaker and sand together is 0.810 J/(g×°C).

The student runs the experiment and collects the data shown in the table below.

Mass of the beaker and sand	1650 g
Mass of butane burner before combustion	225.26 g
Mass of butane burner after combustion	218.20 g
Initial temperature of the beaker of sand	20.°C
Final temperature of the beaker of sand	200.°C

b. Calculate the number of moles of butane that was used in the experiment. Report your answer to the appropriate number of significant figures.

The combustion of butane is represented by the equation below.

$$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 5H_2O(g) + 4CO_2(g) \Delta H^{\circ}_{rm} = -2659 \text{kJ/mol}_{rxn}$$

c. Using the balanced equation for the combustion of butane and $\Delta H_{rxn}°$, determine the amount of heat energy produced by the combustion of butane in the experiment.

Chapter 08 Part 2: Enthalpy Change

d. The student claims that some of the heat energy produced by the combustion of butane was lost to the air surrounding the system. Do your answers to parts (a) and (c) support the student's claim? Explain.

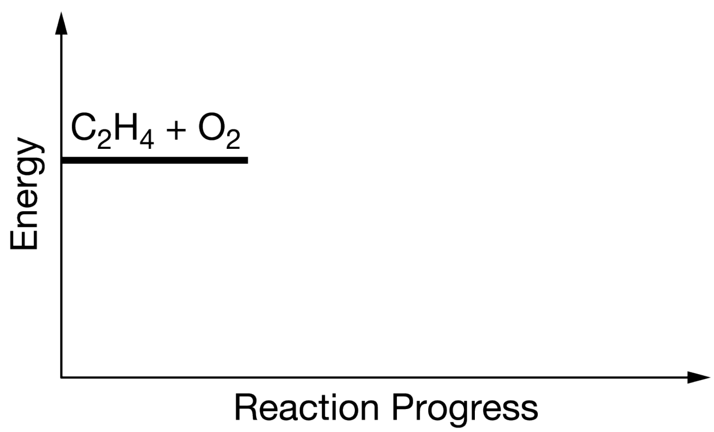
55. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

$C_2H_4(g) + 3\ O_2(g) \rightarrow 2\ CO_2(g) + 2\ H_2O(l) \qquad \Delta H_1^\circ = ?$

The combustion of $C_2H_4(g)$ is represented by the equation above.

(a) Use the enthalpies of formation in the table below to calculate the value of ΔH_1° for the reaction.

	ΔH_f°(kJ/mol)
$C_2H_4(g)$	52
$CO_2(g)$	-394
$H_2O(l)$	-286
$O_2(g)$	0



(b) The energy of the reactants is shown on the energy diagram above. On the right side of the energy diagram, draw a horizontal line segment to indicate the energy of the products. Draw a vertical line segment to indicate ΔH for the reaction, and label it with the correct value.

$C_2H_5OH(l) + 3\ O_2(g) \rightarrow 2\ CO_2(g) + 3\ H_2O(l) \quad \Delta H_2^\circ = -1370\ \text{kJ/mol}_{rxn}$

(c) Based on the information above, identify the ΔH° values that should be added together to determine the value of ΔH_{rxn}° for the following reaction.

$C_2H_4(g) + H_2O(l) \rightarrow C_2H_5OH(l)$

Chapter 08 Part 2: Enthalpy Change

56. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

$$4\,\mathrm{Fe}(s) + 3\,\mathrm{O_2}(g) \rightarrow 2\,\mathrm{Fe_2O_3}(s) \qquad \Delta H^\circ = -1650\,\mathrm{kJ/mol}_{rxn}$$

A student investigates a reaction used in hand warmers, represented above. The student mixes $\mathrm{Fe}(s)$ with a catalyst and sand in a small open container. The student measures the temperature of the mixture as the reaction proceeds. The data are given in the following table.

Time (min)	Temperature of Mixture $(^\circ\mathrm{C})$
0	22.0
1	25.1
2	34.6
3	37.3
4	39.7
5	39.4

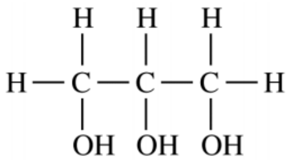
(a) The mixture ($\mathrm{Fe}(s)$, catalyst, and sand) has a total mass of $15.0\ \mathrm{g}$ and a specific heat capacity of $0.72\ \mathrm{J/(g\cdot {}^\circ C)}$. Calculate the amount of heat absorbed by the mixture from 0 minutes to 4 minutes.

(b) Calculate the mass of $\mathrm{Fe}(s)$, in grams, that reacted to generate the amount of heat calculated in part (a).

(c) In a second experiment, the student uses twice the mass of iron as that calculated in part (b) but the same mass of sand as in the first experiment. Would the maximum temperature reached in the second experiment be greater than, less than, or equal to the maximum temperature in the first experiment? Justify your answer.

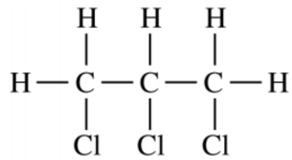
Chapter 08 Part 2: Enthalpy Change

57.



Glycerol

Boiling point 290°C



Trichloropropane

Boiling point 157°C

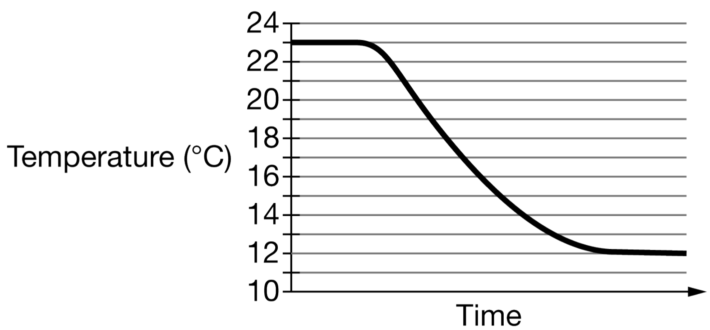
The structural formulas of glycerol and trichloropropane are given above. Both compounds are liquids at 25°C.

a. For each compound, identify all types of intermolecular forces present in the liquid. Explain why glycerol has the higher boiling point in terms of the relative strengths of the intermolecular forces.

b. Glycerol (molar mass 92.09 g/mol) has been suggested for use as an alternative fuel. The enthalpy of combustion, ΔH°_{comb} , of glycerol is -1654 kJ/mol. What mass of glycerol would need to be combusted to heat 500.0 g of water from 20.0°C to 100.0°C ? (The specific heat capacity of water is 4.184 J/(g·°C). Assume that all the heat released by the combustion reaction is absorbed by the water.)

Chapter 08 Part 2: Enthalpy Change

58. **For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**



A student conducts an experiment to determine the value of ΔH°_{soln} for the dissolution of $NaC_2H_3O_2(s)$. The student dissolves 10.0 g of $NaC_2H_3O_2(s)$ in room-temperature water in a beaker and measures the temperature over time. The data are given in the graph above.

(a) The student touches the side of the beaker after the dissolution has occurred and observes that it is cold. What experimental evidence is consistent with the student's observation?

(b) Is the dissolution of $NaC_2H_3O_2(s)$ endothermic or exothermic? Justify your answer in terms of the flow of energy between the system and surroundings.

(c) The student calculated the energy change for the dissolution to be 4600 J. Based on this value, calculate ΔH°_{soln} in $\mathrm{kJ/mol}$.

Chapter 08 Part 2: Enthalpy Change

59. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about the compounds NH_2Cl and NCl_3. The Lewis electron-dot diagrams of the two compounds are shown.

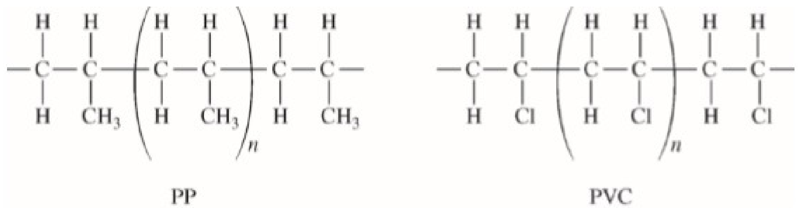
(a) Calculate the number of moles of NH_2Cl (molar mass $51.48\ \mathrm{g/mol}$) present in $1.0\ \mathrm{L}$ of a solution in which the concentration of NH_2Cl is $0.0016\ \mathrm{g/L}$.

(b) NH_2Cl is highly soluble in water, whereas NCl_3 is nearly insoluble. Explain this observation in terms of the types and relative strengths of the intermolecular forces between each of the solutes and water.

(c) The value of $\Delta H^\circ_{vaporization}$ for $NCl_3(l)$ is $32.9\ \mathrm{kJ/mol}$. Calculate the amount of energy required to vaporize a $15.0\ \mathrm{g}$ sample of NCl_3 (molar mass $120.36\ \mathrm{g/mol}$).

Chapter 08 Part 2: Enthalpy Change

60. A student places a mixture of plastic beads consisting of polypropylene (PP) and polyvinyl chloride (PVC) in a 1.0 L beaker containing distilled water. After stirring the contents of the beaker vigorously, the student observes that the beads of one type of plastic sink to the bottom of the beaker and the beads of the other type of plastic float on the water. The chemical structures of PP and PVC are represented by the diagrams below, which show segments of each polymer.



a. Given that the spacing between polymer chains in PP and PVC is similar, the beads that sink are made of which polymer ? Explain.

PP is synthesized from propene, C_3H_6, and PVC is synthesized from vinyl chloride, C_2H_3Cl. The structures of the molecules are shown below.

b. The boiling point of liquid propene (226 K) is lower than the boiling point of liquid vinyl chloride (260 K). Account for this difference in terms of the types and strengths of intermolecular forces present in each liquid.

In a separate experiment, the student measures the enthalpies of combustion of propene and vinyl chloride. The student determines that the combustion of 2.00 mol of vinyl chloride releases 2300 kJ of energy, according to the equation below.

$$2\ C_2H_3Cl(g) + 5\ O_2(g) \rightarrow 4\ CO_2(g) + 2\ H_2O(g) + 2\ HCl(g)\ \Delta H° = -2300\ \text{kJ/mol}_{rxn}$$

c. Using the table of standard enthalpies of formation below, determine whether the combustion of 2.00 mol of propene releases more, less, or the same amount of energy that 2.00 mol of vinyl chloride releases. Justify your answer with a calculation. The balanced equation for the combustion of 2.00 mol of propene is $2\ C_3H_6(g) + 9\ O_2(g) \rightarrow 6\ CO_2(g) + 6\ H_2O(g)$.

Substance	$C_2H_3Cl(g)$	$C_3H_6(g)$	$CO_2(g)$	$H_2O(g)$	$HCl(g)$	$O_2(g)$
Standard Enthalpy of Formation (kJ/mol)	37	21	−394	−242	−92	0

Chapter 08 Part 2: Enthalpy Change

61. **For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

Some building materials contain small capsules filled with paraffin wax to improve the insulating properties of the materials. The melting and freezing of paraffin inside the capsules in the material helps to regulate temperature. Paraffin wax begins to melt at $37^\circ\mathrm{C}$.

(a) While the wax is melting, is the net flow of thermal energy from the wax to the surroundings or from the surroundings to the wax? Justify your answer.

(b) Calculate the amount of thermal energy, in kJ, that is required to melt 15.2 grams of solid paraffin wax when the temperature of the surroundings is above the melting point of paraffin. (The molar mass of paraffin is $282.62\ \mathrm{g/mol}$, and its molar heat of fusion is $48.78\ \mathrm{kJ/mol}$.)

(c) When paraffin wax melts, its density decreases. Provide a particle-level explanation of this observation.

Chapter 08 Part 2: Enthalpy Change

62.

Compound Name	Compound Formula	ΔH^{o}_{vap} (kJ mol^{-1})
Propane	$CH_3CH_2CH_3$	19.0
Propanone	CH_3COCH_3	32.0
1-propanol	$CH_3CH_2CH_2OH$	47.3

Using the information in the table above, answer the following questions about organic compounds.

a. For propanone,

   i. draw the complete structural formula (showing all atoms and bonds);

   ii. predict the approximate carbon-to-carbon-to-carbon bond angle.

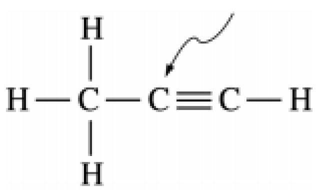
b. For each pair of compounds below, explain why they do not have the same value for their standard heat of vaporization, ΔH°_{vap}, (You must include specific information about <u>both</u> compounds in each pair.)

   i. Propane and propanone

   ii. Propanone and 1-propanol

c. Draw the complete structural formula for an isomer of the molecule you drew in part (a) (i).

d. Given the structural formula for propyne below,



   i. indicate the hybridization of the carbon atom indicated by the arrow in the structure above;

   ii. indicate the total number of sigma (σ) bonds and the total number of pi (◇) bonds in the molecule.